

Lagrangian for an open-chain robotic manipulator

Define frame L_i , attached to and with origin at CoM of link i

$$g_{sl_i}(\theta) = e^{\hat{\xi}_1 \theta_1} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0)$$

Body velocity of frame L_i :

$$V_{sl_i}^b = J_{sl_i}^b(\theta) \dot{\theta} \quad \text{or} \quad V_i^b = J_i(\theta) \dot{\theta}$$

where $J_{sl_i}^b = J_i$ is the body Jacobian corresponding to g_{sl_i}

$$J_i = [\xi_1^\dagger \quad \dots \quad \xi_i^\dagger \quad 0 \quad \dots \quad 0]$$

$$\xi_j^\dagger = \text{Ad}_{\left(e^{\hat{\xi}_j \theta_j} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0) \right)}^{-1} \xi_j \quad j \leq i$$

(In some sense similar to eq. (3.55) p. 117)

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$$\xi_j^\dagger = \text{Ad}_{\left(e^{\hat{\xi}_j \theta_j} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0)\right)}^{-1} \xi_j \quad j \leq i$$

This is as stated in MLS on p. 168. I think that this is incorrect, and should be

$$\xi_j^\dagger = \begin{cases} \text{Ad}_{\left(e^{\hat{\xi}_{j+1} \theta_{j+1}} \dots e^{\hat{\xi}_i \theta_i} g_{sl_i}(0)\right)}^{-1} \xi_j & j < i \\ \text{Ad}_{\left(g_{sl_i}(0)\right)}^{-1} \xi_j & j = i \end{cases}$$

This is compatible with the way MLS writes $J_i(\theta)$ on p. 176, together with the definition of A_{ij} in eq. (4.27) just above, while the above statement on $\xi_j^\dagger$ on p. 168 seems incompatible with p. 176.

Lagrangian for an open-chain robotic manipulator

Kinetic energy of a the i -th link, in terms of its generalized inertia matrix $\mathcal{M}_i$ (see eq. (4.9) p. 162):

$$T_i(\theta, \dot{\theta}) = \frac{1}{2} V_i^{bT} \mathcal{M}_i V_i^b = \frac{1}{2} \dot{\theta}^T J_i^T(\theta) \mathcal{M}_i J_i(\theta) \dot{\theta} \quad (4.17)$$

Total kinetic energy of the manipulator

$$T(\theta, \dot{\theta}) = \sum_{i=1}^n T_i(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M \dot{\theta} \quad (4.18)$$

where the manipulator inertia matrix $M \in \mathbb{R}^{n \times n}$:

$$M(\theta) = \sum_{i=1}^n J_i^T(\theta) \mathcal{M}_i J_i(\theta) \quad (4.19)$$

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Potential energy of a the i-th link

$$V_i(\theta) = m_i g h_i(\theta)$$

Total potential energy of the manipulator

$$V(\theta) = \sum_{i=1}^n V_i(\theta) = \sum_{i=1}^n m_i g h_i(\theta)$$

Lagrangian

$$L(\theta, \dot{\theta}) = \sum_{i=1}^n \left(T_i(\theta, \dot{\theta}) - V_i(\theta) \right) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} - V(\theta)$$

EoM of an open-chain robotic manipulator

$$L(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} - V(\theta) = \frac{1}{2} \sum_{i,j=1}^n M_{ij} \dot{\theta}_i \dot{\theta}_j - V(\theta) \quad (4.20)$$

Equations of motion (EoM):

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = \Upsilon_i, \quad i = 1, 2, \dots, n$$

where Υ_i (capital upsilon) is the actuator torque and other nonconservative generalized forces acting on the i -th joint.

Consider now 1st term in EoM. Only 1st term in (4.20) contributes.

EoM of an open-chain robotic manipulator

$$L(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} - V(\theta) = \frac{1}{2} \sum_{i,j=1}^n M_{ij} \dot{\theta}_i \dot{\theta}_j - V(\theta) \quad (4.20)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = \Upsilon_i, \quad i = 1, 2, \dots, n$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} &= \frac{d}{dt} \left(\sum_{j=1}^n M_{ij} \dot{\theta}_j \right) \\ &= \sum_{j=1}^n \left(M_{ij}(\theta) \ddot{\theta}_j + \dot{M}_{ij}(\theta) \dot{\theta}_j \right) \\ &= \sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \frac{\partial M_{ij}(\theta)}{\partial \theta_k} \dot{\theta}_j \dot{\theta}_k \end{aligned}$$

EoM of an open-chain robotic manipulator

$$L(\theta, \dot{\theta}) = \frac{1}{2} \dot{\theta}^T M(\theta) \dot{\theta} - V(\theta) = \frac{1}{2} \sum_{i,j=1}^n M_{ij} \dot{\theta}_i \dot{\theta}_j - V(\theta) \quad (4.20)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i} = \Upsilon_i, \quad i = 1, 2, \dots, n$$

$$\frac{\partial L}{\partial \theta_i} = \frac{1}{2} \frac{\partial M_{kj}}{\partial \theta_i} \dot{\theta}_k \dot{\theta}_j - \frac{\partial V}{\partial \theta_i}$$

Therefore, EoM (for $i = 1, 2, \dots, n$):

$$\sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \frac{\partial M_{ij}(\theta)}{\partial \theta_k} \dot{\theta}_j \dot{\theta}_k - \frac{1}{2} \frac{\partial M_{kj}}{\partial \theta_i} \dot{\theta}_k \dot{\theta}_j + \frac{\partial V}{\partial \theta_i} = \Upsilon_i$$

EoM of an open-chain robotic manipulator

EoM (for $i = 1, 2 \dots n$):

$$\sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \frac{\partial M_{ij}}{\partial \theta_k} \dot{\theta}_j \dot{\theta}_k - \frac{1}{2} \frac{\partial M_{kj}}{\partial \theta_i} \dot{\theta}_k \dot{\theta}_j + \frac{\partial V}{\partial \theta_i} = \tau_i$$

Define Christoffel symbols

$$\Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial M_{ij}(\theta)}{\partial \theta_k} + \frac{\partial M_{ik}(\theta)}{\partial \theta_j} - \frac{\partial M_{kj}(\theta)}{\partial \theta_i} \right) \quad (4.22)$$

Then

$$\sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \Gamma_{ijk} \dot{\theta}_j \dot{\theta}_k + \frac{\partial V}{\partial \theta_i} = \tau_i \quad (4.21)$$

EoM of an open-chain robotic manipulator

$$\sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \Gamma_{ijk} \dot{\theta}_j \dot{\theta}_k + \frac{\partial V}{\partial \theta_i} = \Upsilon_i \quad (4.21)$$

This is a non-linear 2nd order differential equation in θ , four terms:

- ▶ Inertial forces
- ▶ Centrifugal ($\dot{\theta}_i^2$) and Coriolis ($\dot{\theta}_i \dot{\theta}_j, j \neq i$) forces
- ▶ Potential forces
- ▶ external forces Υ_i

Let external forces be divided into two subcomponents, torque τ_i applied at joint i and “the rest”.

EoM of an open-chain robotic manipulator

Let external forces be divided into two subcomponents, torque τ_i applied at joint i and “the rest”.

Combine “the rest” with the negative of the potential forces to define $-N_i$:

$$-N_i(\theta, \dot{\theta}) = -\frac{\partial V}{\partial \theta_i} - \beta_i \dot{\theta}_i ,$$

where β_i is the viscous damping model of the bearing friction at joint i .

Also define matrix $C(\theta, \dot{\theta}) \in \mathbb{R}^{n \times n}$ (giving (4.23):

$$C_{ij}(\theta, \dot{\theta}) = \sum_{j,k=1}^n \Gamma_{ijk} \dot{\theta}_k = \frac{1}{2} \left(\frac{\partial M_{ij}(\theta)}{\partial \theta_k} + \frac{\partial M_{ik}(\theta)}{\partial \theta_j} - \frac{\partial M_{kj}(\theta)}{\partial \theta_i} \right) \dot{\theta}_k$$

EoM of an open-chain robotic manipulator

$$\sum_{j=1}^n M_{ij} \ddot{\theta}_j + \sum_{j,k=1}^n \Gamma_{ijk} \dot{\theta}_j \dot{\theta}_k + \frac{\partial V}{\partial \theta_i} = \tau_i \quad (4.21)$$

Using newly defined C and N matrices and joint actuator torque vector τ :

$$M(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + N(\theta, \dot{\theta}) = \tau \quad (4.24)$$

Proposition 4.3 p. 176:

$$M(\theta) = \sum_{i=1}^n J_i^T(\theta) \mathcal{M}_i J_i(\theta) \quad (4.19)$$

$$J_i(\theta) = \text{Ad}_{g_{sl_i(0)}^{-1}} \begin{bmatrix} A_{i1}\xi_1 & \cdots & A_{ii}\xi_i & 0 & \cdots & 0 \end{bmatrix}$$

Inertia matrix of i -th link reflected into S (p. 176):

$$\mathcal{M}'_i = \text{Ad}_{g_{sl_i(0)}^{-1}}^T \mathcal{M}_i \text{Ad}_{g_{sl_i(0)}^{-1}} \quad (4.28)$$

Proposition 4.3 p. 176:

$$M_{ij}(\theta) = \sum_{l=\max(i,j)}^n \xi_i^T A_{li}^T \mathcal{M}'_l A_{lj} \xi_j \quad (4.29)$$

Proposition 4.3 p. 176:

$$M_{ij}(\theta) = \sum_{l=\max(i,j)}^n \xi_i^T A_{li}^T \mathcal{M}'_l A_{lj} \xi_j \quad (4.29)$$

$$\frac{\partial M_{ij}}{\partial \theta_k} = \sum_{l=\max(i,j)}^n \left[\frac{\partial(\xi_i^T A_{li}^T)}{\partial \theta_k} \mathcal{M}'_l A_{lj} \xi_j + \xi_i^T A_{li}^T \mathcal{M}'_l \frac{\partial(A_{lj} \xi_j)}{\partial \theta_k} \right]$$

If we can show that

$$\frac{\partial(A_{lj} \xi_j)}{\partial \theta_k} = A_{lk} [A_{kj} \xi_j, \xi_l]$$

we would have proved eq. (4.30):

$$\frac{\partial M_{ij}}{\partial \theta_k} = \sum_{l=\max(i,j)}^n \left([A_{ki} \xi_i, \xi_l]^T A_{lk}^T \mathcal{M}'_l A_{lj} \xi_j + \xi_i^T A_{li}^T \mathcal{M}'_l A_{lk} [A_{kj} \xi_j, \xi_l] \right)$$

Proposition 4.3 p. 177:

For $i > j$:

$$g_{ji} = e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_i\theta_i}$$

Then

$$g_{ij} = g_{ji}^{-1} = e^{-\hat{\xi}_i\theta_i} \dots e^{-\hat{\xi}_{j+1}\theta_{j+1}}$$

For $i \geq j$:

$$g_{ij} = \begin{cases} e^{-\hat{\xi}_i\theta_i} \dots e^{-\hat{\xi}_{j+1}\theta_{j+1}} & i > j \\ I & i = j \end{cases}$$

Proposition 4.3 p. 177:

For $i > j$:

$$g_{ji} = e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_i\theta_i}$$

$$g_{ij} = e^{-\hat{\xi}_i\theta_i} \dots e^{-\hat{\xi}_{j+1}\theta_{j+1}}$$

Eq. (4.27) p. 176:

$$A_{ij} = \text{Ad}_{(e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_i\theta_i})}^{-1} = \text{Ad}_{g_{ji}}^{-1} = \text{Ad}_{g_{ij}}$$

For k an integer $i \geq k \geq j$: $g_{ij} = g_{ik}g_{kj}$

Proposition 4.3 p. 177:

For $i \geq k \geq j$ (using Lemma 2.13 p. 56 repeatedly):

$$\frac{\partial(A_{lj}\xi_j)}{\partial\theta_k} = \left(\frac{\partial}{\partial\theta_k} \left(g_{lj}\hat{\xi}_j g_{lj}^{-1} \right) \right)^\vee = \left(\frac{\partial g_{lj}}{\partial\theta_k} \hat{\xi}_j g_{lj}^{-1} + g_{lj} \hat{\xi}_j \frac{\partial g_{lj}^{-1}}{\partial\theta_k} \right)^\vee$$

$$g_{lj} = e^{-\hat{\xi}_l\theta_l} \dots e^{-\hat{\xi}_{j+1}\theta_{j+1}} \quad g_{lj}^{-1} = e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_l\theta_l}$$

$$\frac{\partial g_{lj}}{\partial\theta_k} = -e^{-\hat{\xi}_l\theta_l} \dots \hat{\xi}_k e^{-\hat{\xi}_k\theta_k} \dots e^{-\hat{\xi}_{j+1}\theta_{j+1}} = -g_{lk} \hat{\xi}_k g_{kj}$$

$$\frac{\partial g_{lj}^{-1}}{\partial\theta_k} = e^{\hat{\xi}_{j+1}\theta_{j+1}} \dots e^{\hat{\xi}_k\theta_k} \hat{\xi}_k \dots e^{\hat{\xi}_l\theta_l} = g_{kj}^{-1} \hat{\xi}_k g_{lk}^{-1}$$

Proposition 4.3 p. 177:

For $i \geq k \geq j$ (using Lemma 2.13 p. 56 repeatedly):

$$\begin{aligned}
 \frac{\partial(A_{lj}\xi_j)}{\partial\theta_k} &= \left(\frac{\partial}{\partial\theta_k} \left(g_{lj}\hat{\xi}_j g_{lj}^{-1} \right) \right)^\vee = \left(\frac{\partial g_{lj}}{\partial\theta_k} \hat{\xi}_j g_{lj}^{-1} + g_{lj} \hat{\xi}_j \frac{\partial g_{lj}^{-1}}{\partial\theta_k} \right)^\vee \\
 &= \left(-g_{lk} \hat{\xi}_k g_{kj} \hat{\xi}_j g_{lj}^{-1} + g_{lj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k g_{lk}^{-1} \right)^\vee \\
 &= \left(-g_{lk} \hat{\xi}_k g_{kj} \hat{\xi}_j g_{kj}^{-1} g_{lk}^{-1} + g_{lk} g_{kj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k g_{lk}^{-1} \right)^\vee \\
 &= \left(g_{lk} \left(-\hat{\xi}_k g_{kj} \hat{\xi}_j g_{kj}^{-1} + g_{kj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k \right) g_{lk}^{-1} \right)^\vee \\
 &= \text{Ad}_{g_{lk}} \left(-\hat{\xi}_k g_{kj} \hat{\xi}_j g_{kj}^{-1} + g_{kj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k \right)^\vee
 \end{aligned}$$

Proposition 4.3 p. 177:

$$\begin{aligned}\frac{\partial(A_{lj}\xi_j)}{\partial\theta_k} &= \text{Ad}_{g_{lk}} \left(-\hat{\xi}_k g_{kj} \hat{\xi}_j g_{kj}^{-1} + g_{kj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k \right)^\vee \\ &= A_{lk} \left(-\hat{\xi}_k g_{kj} \hat{\xi}_j g_{kj}^{-1} + g_{kj} \hat{\xi}_j g_{kj}^{-1} \hat{\xi}_k \right)^\vee \\ &= A_{lk} [A_{kj} \xi_j, \xi_k] \quad (\text{using Lie brackets definition})\end{aligned}$$

QED

For $l \geq k \geq j$:

$$g_{lj} = g_{lk}g_{kj}$$

and

$$g_{lj}^{-1} = (g_{lk}g_{kj})^{-1} = g_{kj}^{-1}g_{lk}^{-1}$$